Shashank Shaga

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Education

University of Florida — Gainesville, FL
B.S Computer Science, AI certificate. GPA: 3.85 — **May 2028**
- 3x Dean's list, FAS merit scholarship, Coursework: Data Structures & Algorithms

Technical Skills

Languages: Python, C++, Swift, Dart, Java, JavaScript, Typescript, SQL
Frameworks: React, Flutter, Next.js, SwiftUI, Flask, FastAPI
Developer Tools: Git, MongoDB, Firebase, Snowflake, Figma, LangChain, NPM, Claude Code
Libraries: PyTorch, OpenCV, MediaPipe, NumPy, pandas, Matplotlib, LangChain

Experience

GE Appliances — Aug 2026 – Present
Software Engineering Co-op — **Louisville, KY**
- Engineered the migration of **SmartHq's 1M+ daily active user** app from native IOS using swift (Viper architecture) to Flutter implementing **10+ production pages**, contributing to a **30% reduction** in bugs.
- Developed connected appliance experiences by integrating Flutter applications applying dart with **50+ backend API endpoints**, WebSocket communication and Firebase analytics services for **15 appliances**
- Resolved **10+ production tickets** and delivered application enhancements across a large-scale IoT mobile and enhancing enterprise CI/CD pipeline by adding 3 major build tests in Git.

Gator AI — Jan 2026 – May 2026
Machine Learning Engineer — **Gainesville, FL**
- Built a high-fidelity chaotic dynamics simulation of a double pendulum using NumPy), generated **6M+ trajectory data** points to train and validate neural networks using **PyTorch**.
- Harnessed HiPerGator (supercomputer) cluster to construct chaos maps and integrating 20 frames from sql to detect physical motion such as **Missile detection.**
- Architected neural networks to approximate chaotic state transition **reducing prediction error by 35%** compared to baseline numerical extrapolation (numpy) and enabling scalable AI predicted chaos forecasting.

Machine Learning Laboratory — Aug 2025 – Jan 2026
Software Engineer — **Gainesville, FL**
- Advanced underwater robot navigation efficiency by targeting a **30% reduction** in travel distance by deploying ROS-based path optimization utilizing C++.
- Tested underwater computer vision models to improve RGB spectrum recognition in challenging aquatic environments with rigorous unit tests to ensure **95 % autonomous route accuracy**.
- Led the development of Navigator website redesign using **Flask, React + JS**, integrating **Figma designs**.

Projects

Chaos Wall | *PyTorch, NumPy, MongoDB, pandas, LangChain, RestAPI* — Jan 2026 – Jul 2026
- Developed a physics-informed machine learning framework for chaotic system prediction by combining **LSTM state encoder** with a **Hamiltonian Neural Network (HNN)** to model double pendulum dynamics, enforcing energy conservation through learned Hamiltonian functions and autograd-based dynamics.
- Generated and processed 6M+ sequential training windows from **500 simulated trajectories** stored in **Parquet format**; trained PyTorch LSTM and HNN models to compare data-driven vs physics-constrained forecasting,

Visi Blind | ***(Best Service Project)*** *- Python, React, MediaPipe, MongoDB, OpenCV* — Mar 2026 – May 2026
- Built **full-stack AI accessibility** platform using React, TypeScript, Python, and OpenCV for real-time gesture.
- Integrated AI models (MediaPipe, Gemini, ElevenLabs) with MongoDB vector storage to deliver voice-based environmental understanding.

Involvement

Achievements: 2x Hackathon winner, Winner GatorAI 2026, 9 years in competitive dance, 1900 Chess rating.
Clubs: GatorAI (Project Lead), Society of Asian engineers and scientists (SASE), ColorStack